Basic Principles of Medicine 1

Module: Foundation to Medicine
Lecture: FTM 44

Multifactorial Inheritance

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Acknowledgement
Dr. Mary Maj
Dr. Sharmila Upadhya

Objectives

SOM.MK.III.BPM1.1.FTM.3.GNET.0115	Analyze how quantitative traits can be controlled by many genes each of which might have minimal contribution to the phenotype
SOM.MK.III.BPM1.1.FTM.3.GNET.0116	Discuss general concepts of multifactorial disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0117	Analyze the threshold model of multifactorial inheritance
SOM.MK.III.BPM1.1.FTM.3.GNET.0118	Analyze heritability of multifactorial disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0119	Discuss methods used to identify genetic or environmental risk factors and contributors for multifactorial disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0114	Discuss quantitative traits in terms of multifactorial inheritance
SOM.MK.III.BPM1.1.FTM.3.GNET.0115	Analyze how quantitative traits can be controlled by many genes each of which might have minimal contribution to the phenotype

Recommended Reading

Complex Inheritance of Common Multifactorial Disorders:

Thompson & Thompson Genetics and Genomics in Medicine, Chapter 9, 149-173; https://www.clinicalkey.com/student/content/book/3-s2.0-B978032354762800009X

- Found through ClinicalKey link on your Course Sakai Site

Korf & Irons Multifactorial Inheritance, Chapter 5

- Found through your Course Sakai Site: General information: Book Access: Genetics Book Chapters Required and Recommended Readings

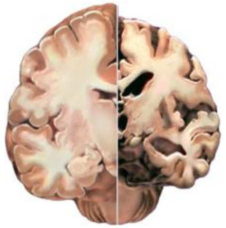
Majority of human genetic disorders are multifactorial because they are **Polygenic**

Important disorders to know:

- **Pyloric stenosis:** progressively worsening projectile vomiting, failure to thrive and dehydration

- **Cleft lip:** surgical correction of a developmental abnormality improves the quality of life for a child

- **Alzheimer disease:** Age is the most important risk factor



- **Multiple genes and the environment:** heart disease, cancer, autism, asthma, type 2 diabetes

Majority of human genetic disorders are multifactorial

Multifactorial polygenic trait

- **Genetic Variance**
  - Recessive alleles have additive effects (many)
  - Dominant alleles have additive effects (fewer)
  - Modifier genes can influence positively or negatively (epistasis)
- **Environmental Variance**
  - Nutrition (utero vs. childhood vs. adulthood)
  - Environmental (toxins, climate)
  - Lifestyle choices (smoking, exercise, stress levels)
  - Socioeconomic status, access to healthcare

Disorders that show multifactorial inheritance	Examples
Congenital anomalies	**Cleft lip & palate**, **spina bifida**, **pyloric stenosis**, club foot, congenital heart defects
Acquired Traits/Diseases of Childhood & Adult life	
Quantitative traits	Height, weight (obesity), blood pressure
Common metabolic disorders	Hypertension, diabetes mellitus, high cholesterol
Cancers	Breast, ovarian, bowel, prostate, skin cancers
Neurological disorders	**Alzheimer disease**, epilepsy, schizophrenia, bipolar disorder, multiple sclerosis, Parkinson disease, manic depression
Other	Asthma, glaucoma, inflammatory bowel (Crohn disease & ulcerative colitis), rheumatoid arthritis

Multifactorial Inheritance (Complex inheritance)

- Many **traits** tend to cluster within families but do not follow Mendelian laws
  - Family aggregation is seen
  - More common among **close relatives** & less common in distant relatives of the proband
- Combination of genetic and non-genetic factors (e.g., environment & nutrition) to produce phenotype
- Assessment of recurrence risk in families
  - Quantitative traits and Polygenic inheritance (Effect of multiple genes)
  - Genetic liability threshold model
- Measuring genetic contribution:
  - Familial clustering in population without clear Mendelian pattern
  - Monozygotic twin concordance
  - Studies of heritability
- Identifying gene susceptibility loci: GWAS studies
- Role of environmental factors:
  - Population/Migration studies
  - Adoption studies

Quantitative traits and Polygenic Inheritance

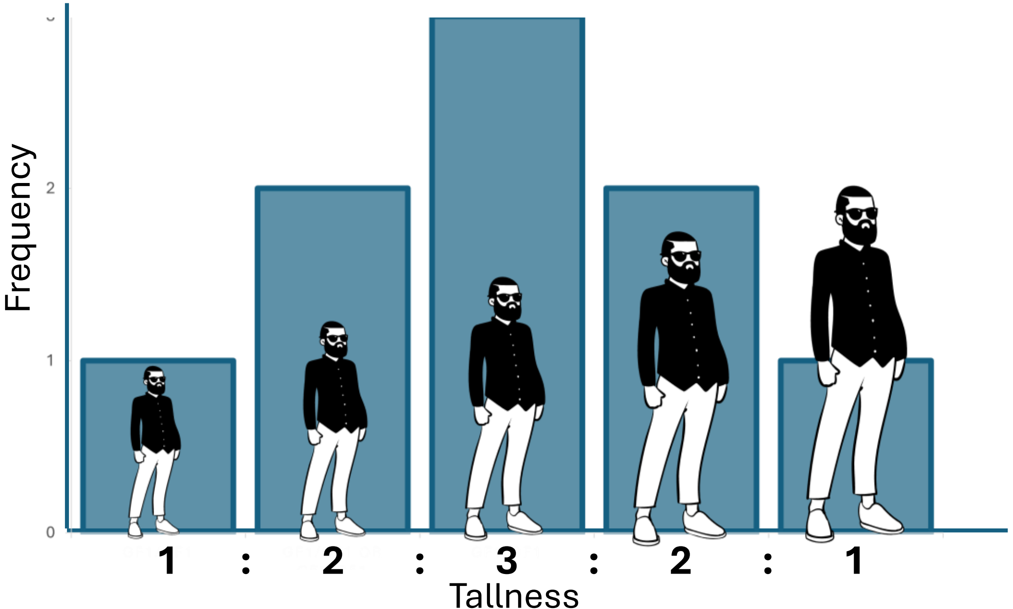
E.g., final adult height

Hypothetical: 1 gene, 2 alleles A and a,
where A=Tall
3 possible phenotypes from 2 heterozygous parents (recall Punnet square):

1	**aa**	(0 dose tall allele)
2	**Aa, aA**	(1 dose tall allele)
3	**AA**	(2 dose tall allele)

Hypothetical: 2 genes, 4 alleles A, a, B, b
where **A** and **B**=Tall alleles
5 possible phenotypes, heterozygous parents at each locus

1	**aabb**	(0 dose tall allele)
2	**Aabb, aaBb**	(1 dose tall allele)
3	**AaBb, AAbb, aaBB**	(2 dose tall allele)
4	**AaBB, AABb**	(3 dose tall allele)
5	**AABB**	(4 dose tall allele)



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Quantitative traits and Polygenic Inheritance

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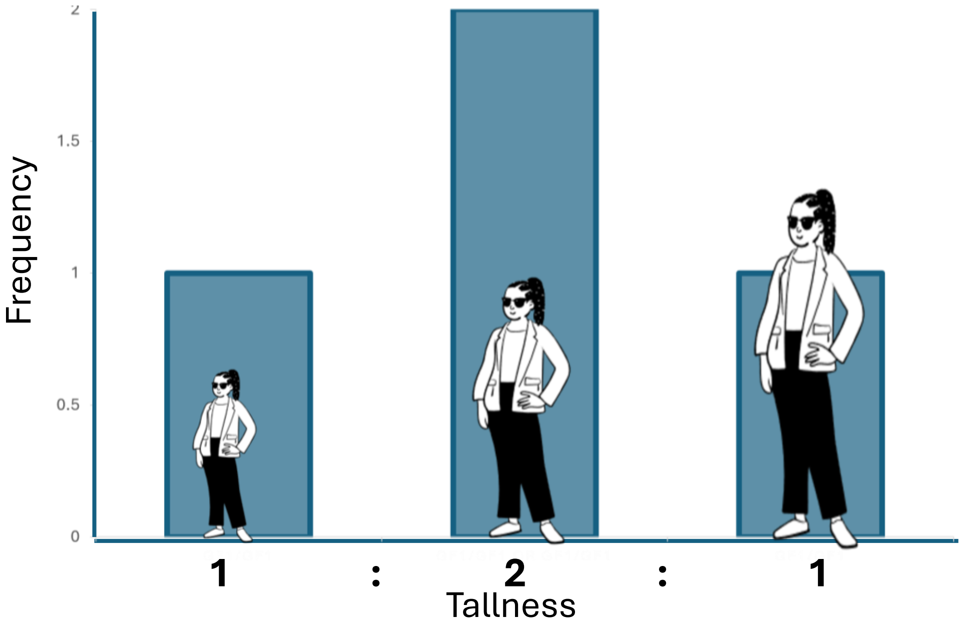
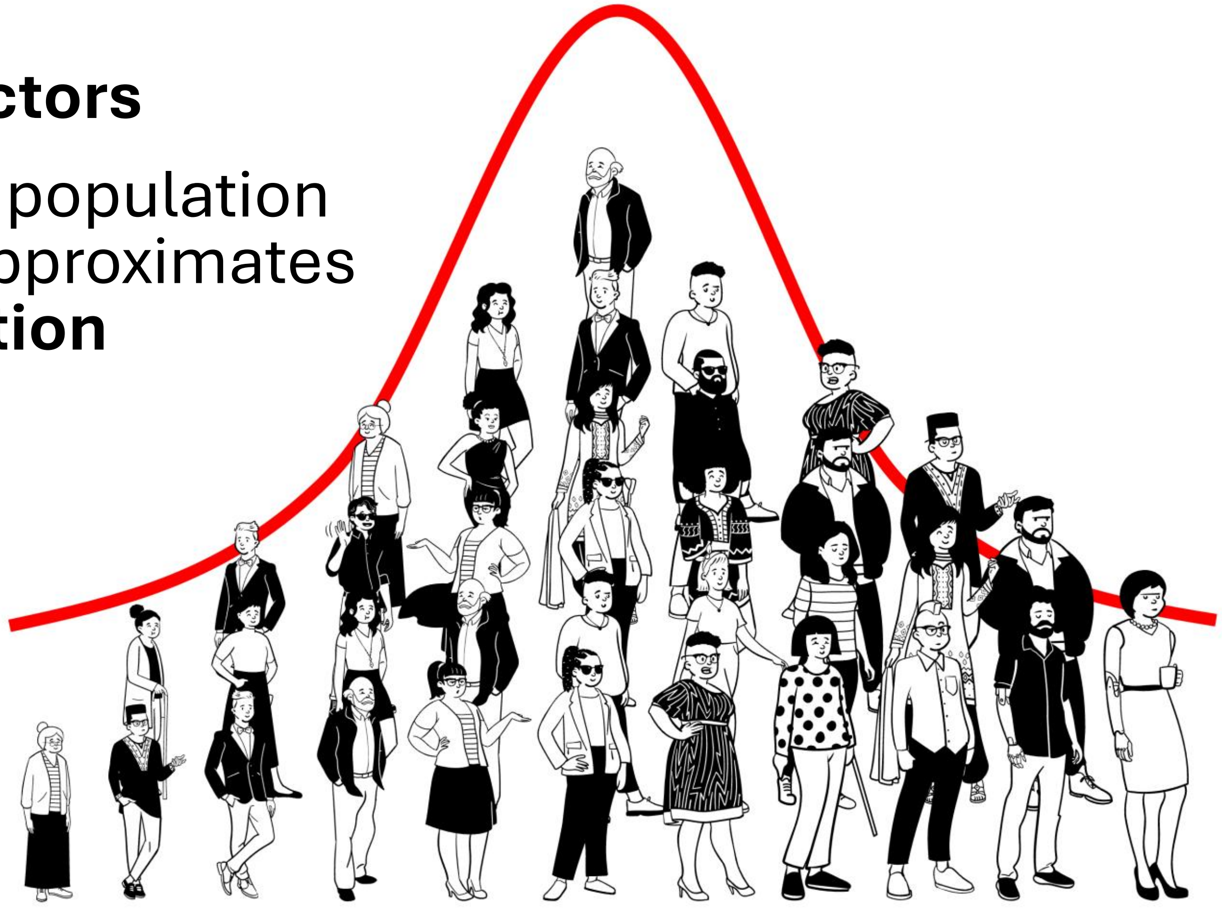


1	**aa**	(0 dose tall allele)
2	**Aa, aA**	(1 dose tall allele)
3	**AA**	(2 doses tall allele)

Quantitative traits and Polygenic Inheritance

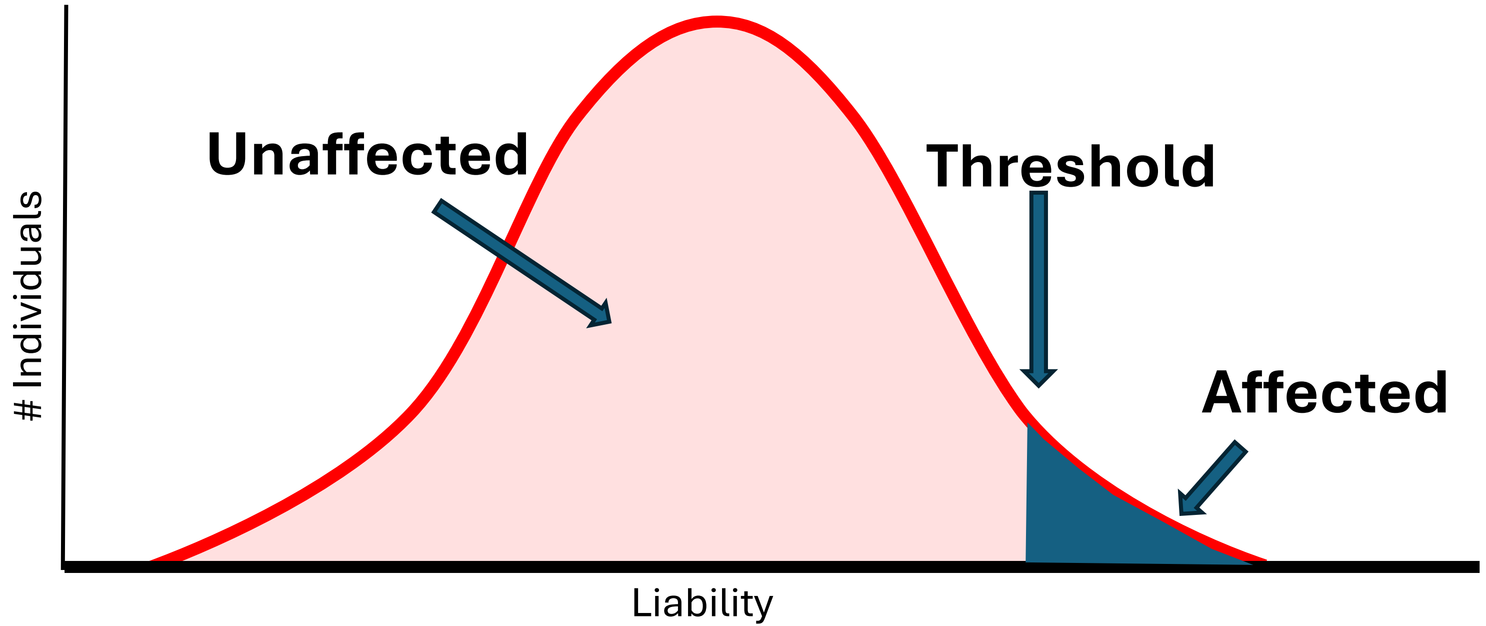
- Measurable phenotypic characteristic with **continuous variation** within a population
- Influenced by:
    - **Multiple genes** & **Environmental factors**
- A graph of the number of individuals in a population following a quantitative value typically approximates **normal/gaussian/bell shaped distribution**
- Typically measured numerically:
    - Height
    - Weight (BMI)
    - Blood pressure
    - Serum cholesterol



Population sorted by Height

Multifactorial disorders: Liability Threshold Model

- Attempts to describe a populations genetic and environmental susceptibility
- Multifactorial inheritance produces a **continuous phenotype**
- Context of **continuous variation** using the liability/threshold model
- All factors (genetic and environmental) that contribute to the liability of disease produce **normal distribution**
- At some point there is enough contribution to an underlying quantitative variable to cause expression of abnormal phenotype = **Threshold**



Where an individual lies in this distribution will be a result of Genes & Environment
(lifestyle, surroundings, nutrition, etc.)

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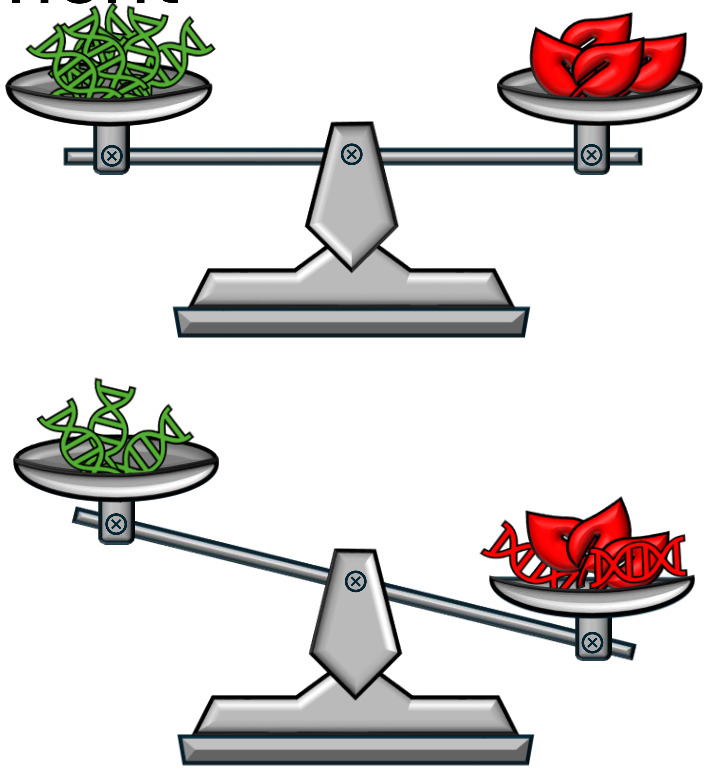
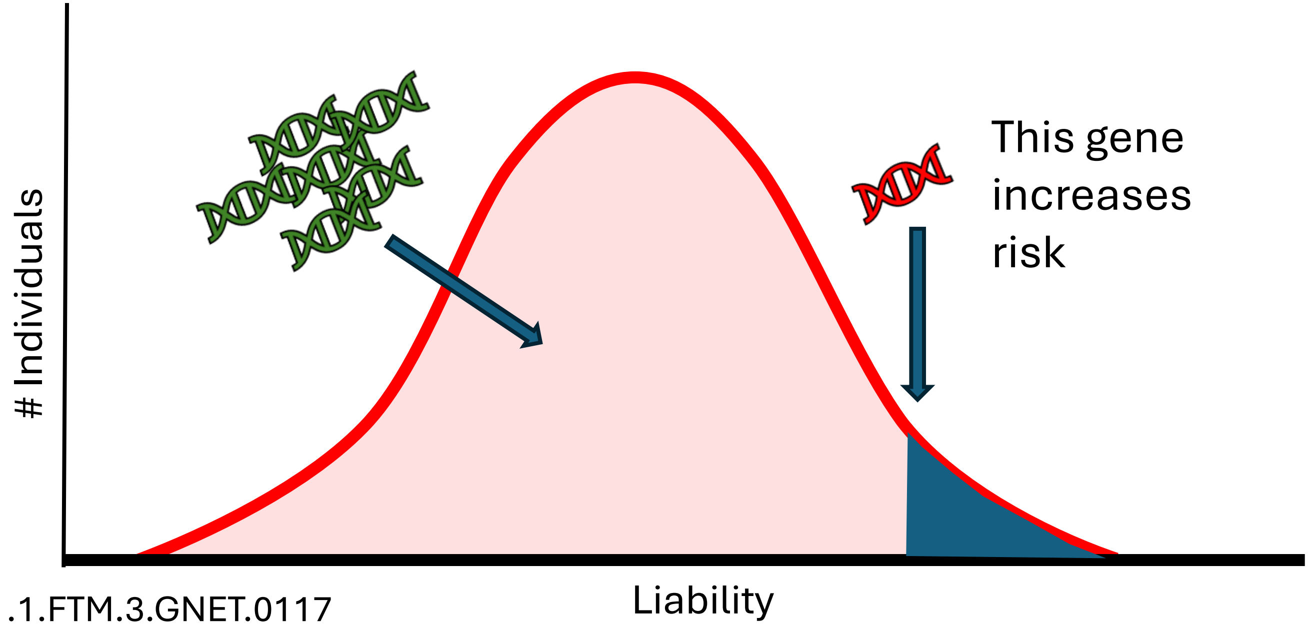
Multifactorial disorders: Liability Threshold Model
Disease determined by both genes & environment

- If you have only “good” genes, you can have a lot of “susceptibility” in the environment and be unaffected

- But if you start out with some “susceptibility” genes, it does not take much “susceptibility” in the environment to tip you over the edge



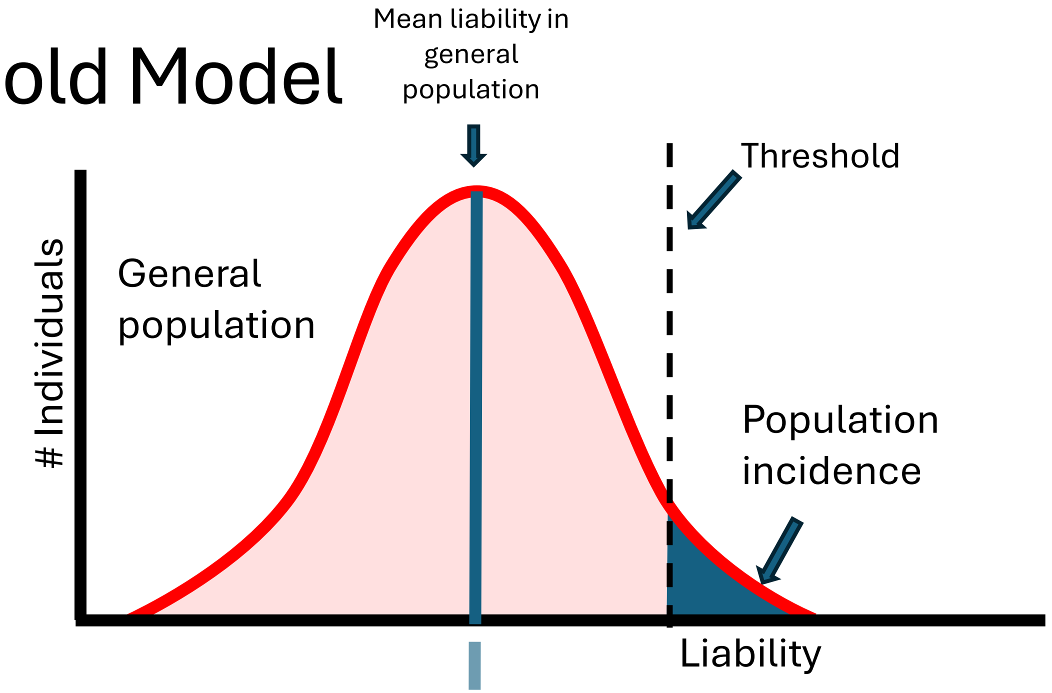


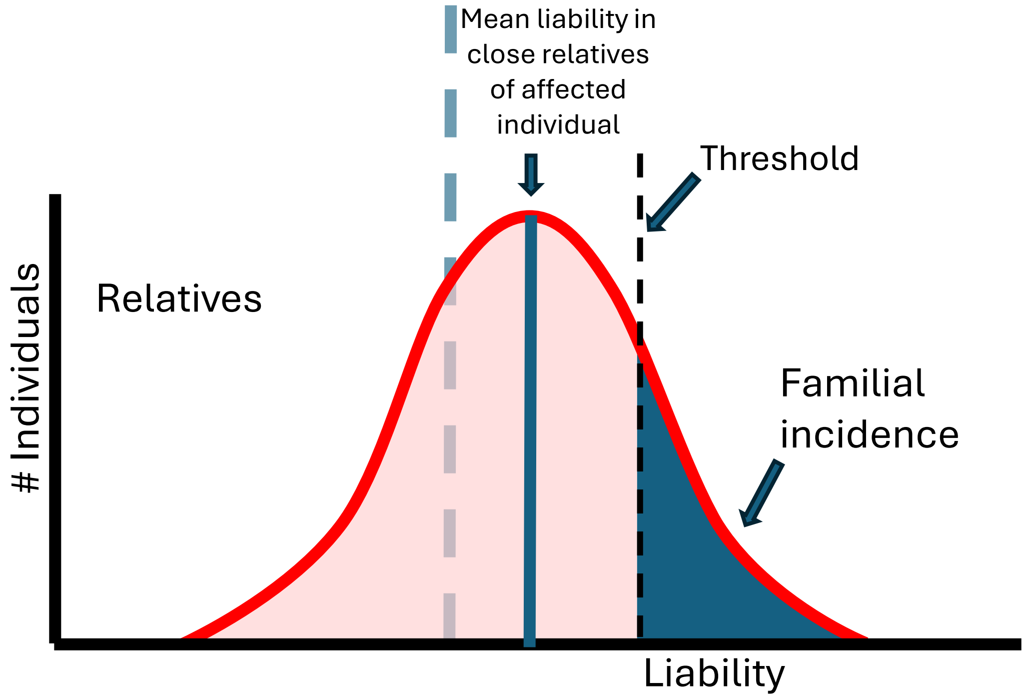
Good gene
Susceptibility gene
Good environment
Susceptibility environment

Multifactorial disorders: Liability Threshold Model

Disease state with familial aggregation

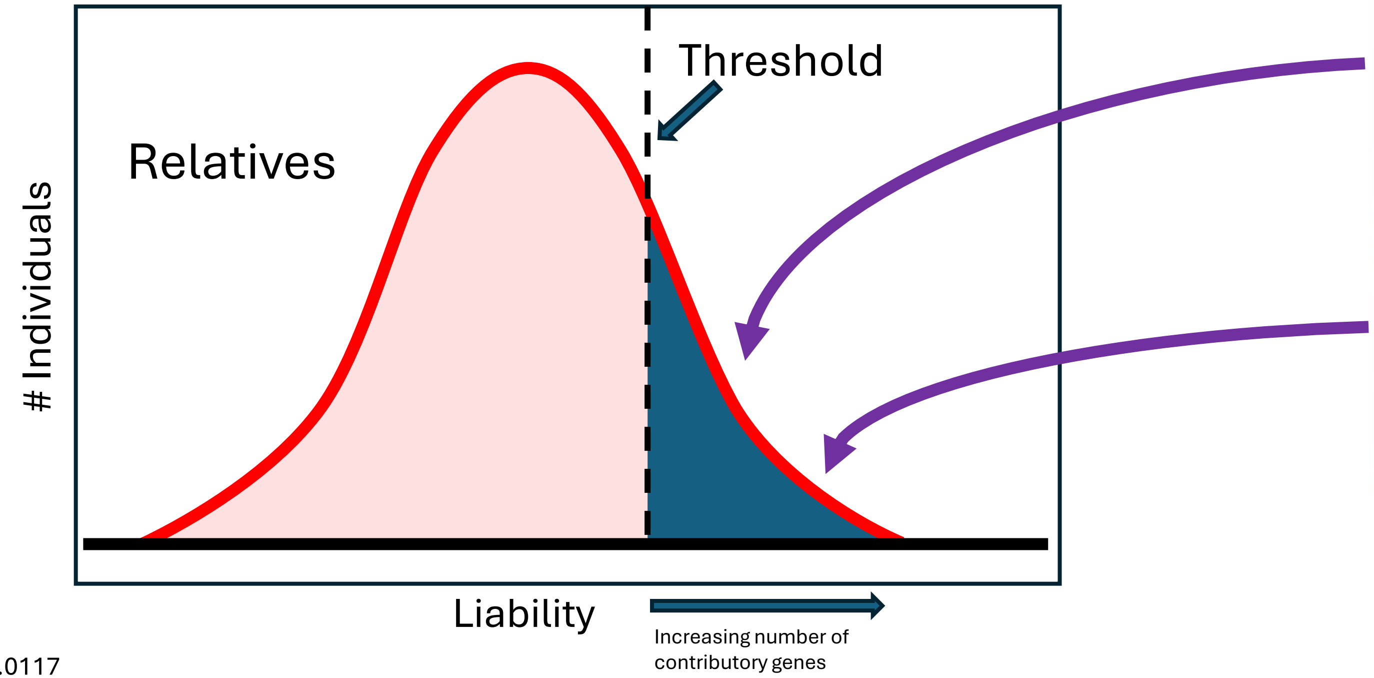
- **A threshold point for general population**
- "Liability" threshold is composed of both genetic and environmental factors
- Families with "bad" or "susceptibility" genes with additive effect, curve moves to the right or the threshold line moves to the left (higher risk)
  - Increased probability of affected family members as they share susceptibility genes
  - **Closer the relation, higher the risk**
  - **Higher liability of the disorder**





Multifactorial disorders: Liability Threshold Model with severity
Example: Cleft lip and palate

- Higher recurrence risk in relatives of **severely affected individuals**
  - Overall incidence is ~1/1000
  - Severely affected patients most likely have **more susceptibility genes** (at the high end of the liability curve)



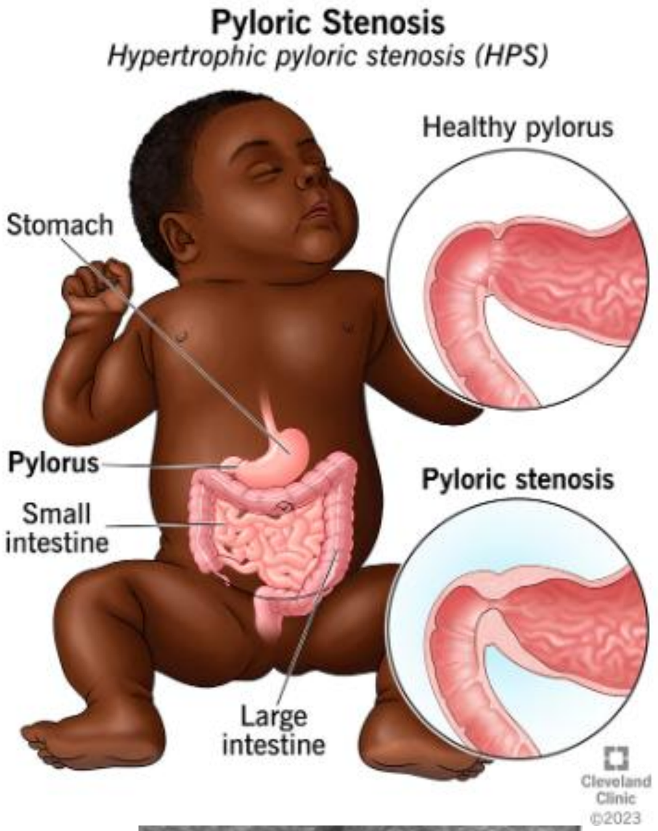
Unilateral cleft lip

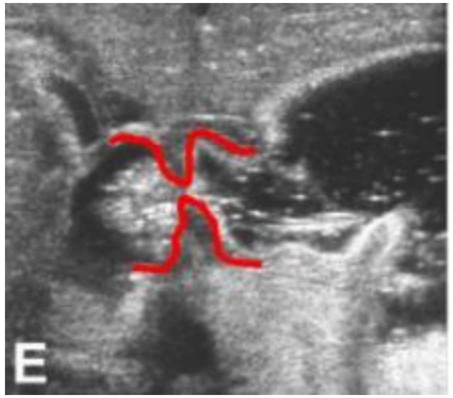
Bilateral cleft lip & palate

Multifactorial disorders: Liability Threshold Model with Bias

Example: Pyloric Stenosis has **sex bias**

- *Pyloric stenosis* = hypertrophy of ***pylorus*** (between stomach and duodenum), causing it to narrow (***stenosis***) impedes gastric emptying
  - Causes severe vomiting in babies
  - Palpation of abdomen may reveal mass in epigastrium
  - Dehydration and acid-base imbalance
  - 15% of infants affected have a positive family history
- **Pyloric stenosis more common in male infants than in female infants**
  - 1/200 for males & 1/1000 for females

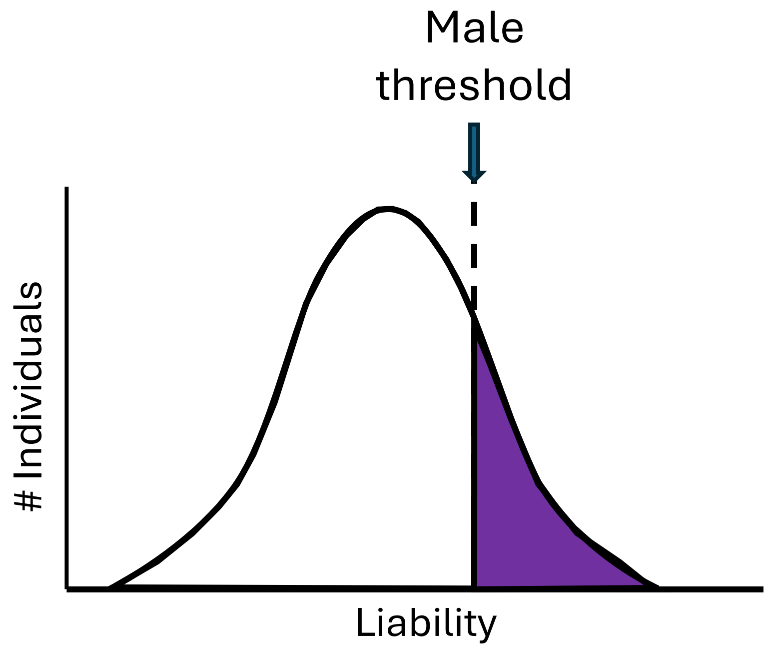


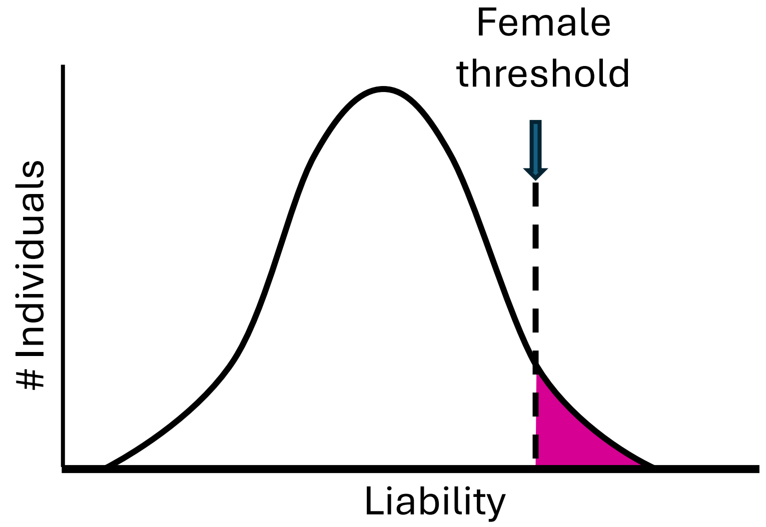


Multifactorial disorders: Liability Threshold Model with Bias

Example: Pyloric Stenosis has **sex bias**

- **Males are more prone** to the condition (have lower liability threshold)
- Therefore, **if a female is affected**, she must be at the high end of the curve (more susceptibility genes)
- **∴ more susceptibility genes!**
  - Children (or siblings) of an affected female are more at risk than children (or siblings) of an affected male
  - Sons (or brothers) of an affected female are more at risk than her daughters (or sisters)





Measuring Genetic Contribution in Multifactorial Disorders

- Family studies
- Twin studies
- Association studies

Measuring Genetic Contribution in Multifactorial Disorders

Example: Family studies

- The closer the relationship to a person with a multifactorial disorder, the higher chance of **sharing contributory alleles**
- = **Familial Relative Risk**
- The risk drops by half for every degree distance from affected person
- Relative risk ratio can be calculated = λ_r

Degrees of Relationship		
First degree (Highest risk)	Parent-offspring	1/2
	Full siblings	
Second degree	Half siblings	1/4
	Uncle-niece/Aunt-nephew	
	Grandparents	
	Grandchildren	
Third degree (Least risk)	First cousins	1/8
	Great grandparents	
	Great grandchildren	

$$\lambda_r = \frac{\text{Prevalence of the disease in the relatives of an affected person}}{\text{Prevalence of the disease in the general population}}$$

We will not be performing the calculation for relative risk (λ_r) in this class

Measuring Genetic Contribution in Multifactorial Disorders

Example: Twin Studies

Strong genetic component

Disorder	Prevalence in population (%)	Heritability (%)
Schizophrenia	1	85
Asthma	4	80
Cleft lip cleft/palate	0.1	76
Pyloric stenosis	0.3	75
Club foot	0.1	68
Coronary artery disease	3	65
Hypertension	5	62
Anencephaly & spina bifida	0.3	60
Peptic ulcer	4	37
Congenital heart disease	0.5	35

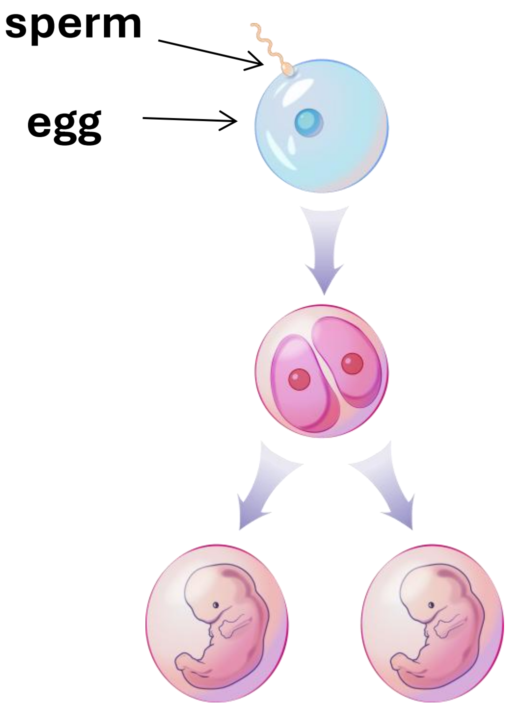
More environmental (exposure to *H. pylori*)

- **% Heritability** is the proportion of variation in a "trait" attributed to inherited factors (genetic factors)
- **0% (**close to zero) indicates that variability in a trait is likely **environmental** and not genetics
  - E.g. language
- **100%** (close to 1) indicates variability in a trait due mainly to **genetics** and little environmental influence
- **Heritability** estimated from twin studies
  - Identical vs. fraternal

Measuring Genetic Contribution in Multifactorial Disorders
Example: Disease concordance in Monozygotic & Dizygotic Twins

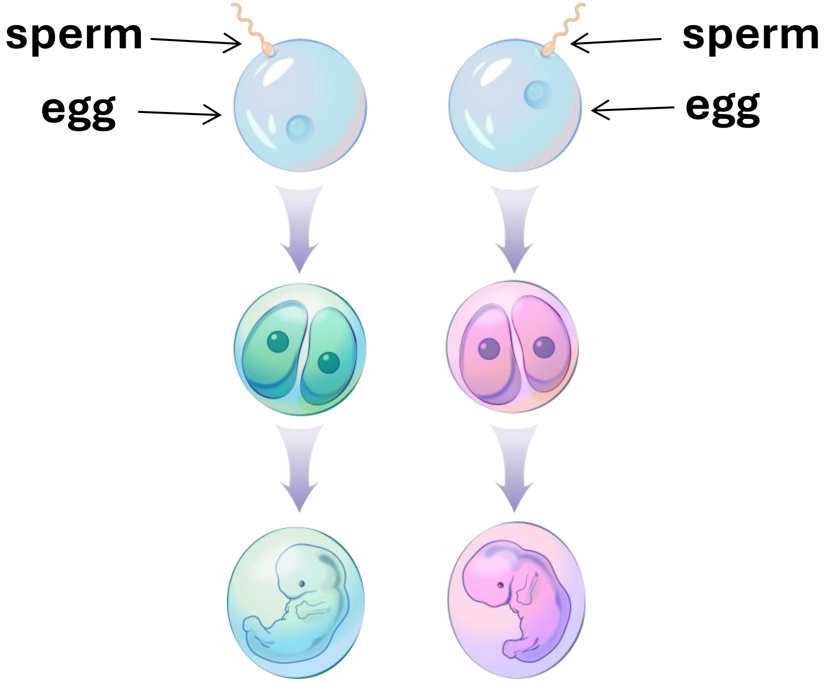
Monozygotic (MZ) twins

- Derived from a single ovum
- Genetically identical and share intrauterine environment



Dizygotic (DZ) twins (controls)

- Derived from two separate ova and sperm
- Siblings (~50% identical) but share intrauterine environment



Measuring Genetic Contribution in Multifactorial Disorders

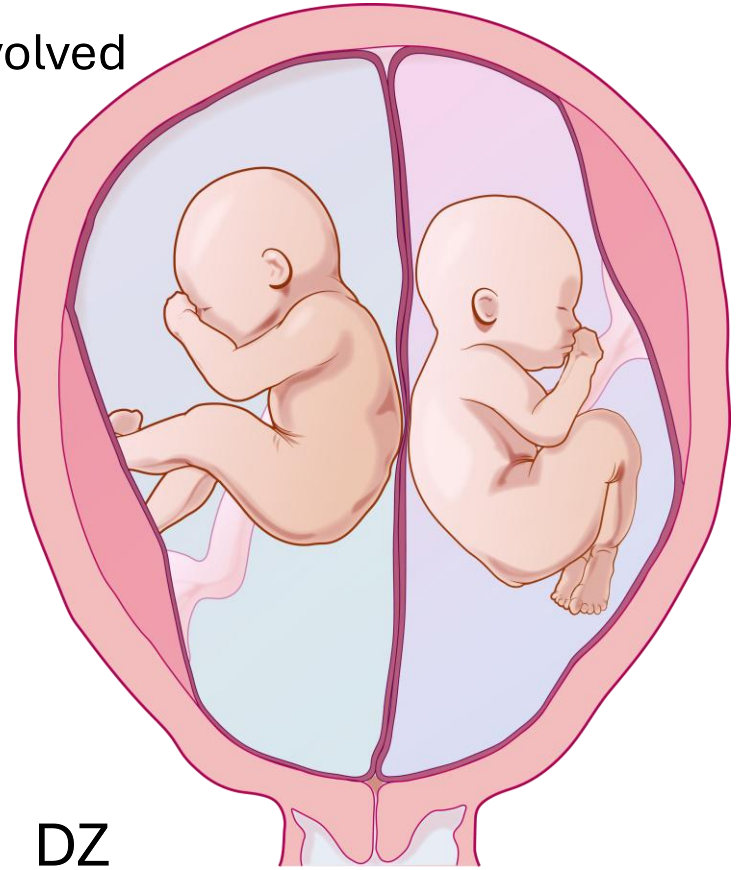
Example: Disease concordance in Monozygotic Twins (MZ)

- **Concordance** means both twins have the disorder
- **Discordance** means only one twin is affected
- In MZ twins:
  - If concordance is ~100%, it is a **genetically** determined trait
  - If concordance is <100%, **environmental factors** involved

MZ

The greater the difference in concordance between MZ and DZ twins = greater genetic input

MZ>>DZ



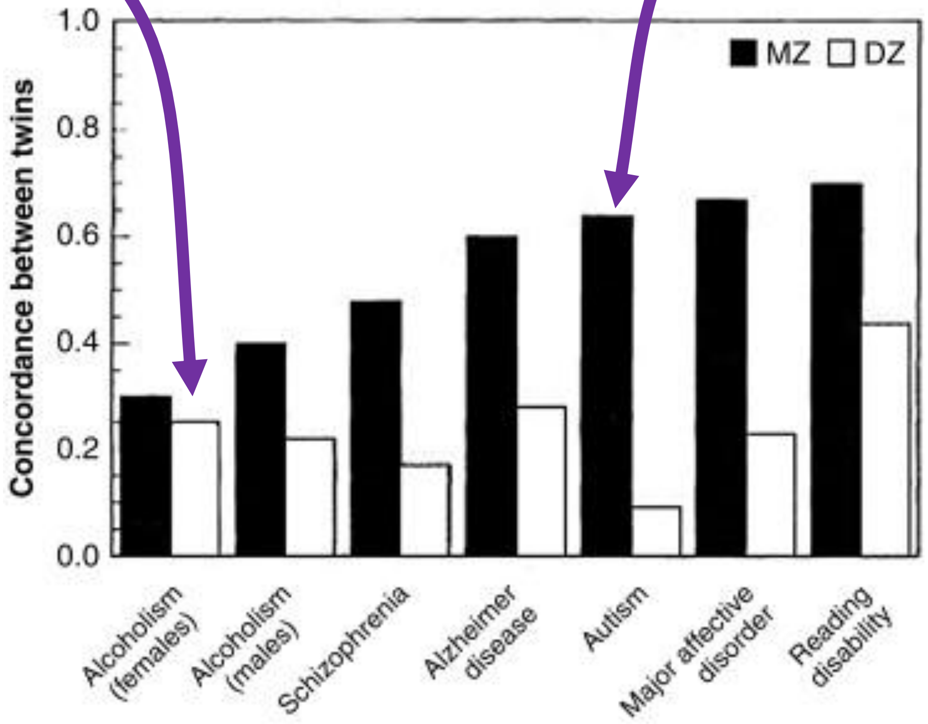
DZ

Measuring Genetic Contribution in Multifactorial Disorders

Example: Disease concordance in Monozygotic (MZ) vs Dizygotic (DZ) twins

Concordance similar between MZ and DZ shows **environment** plays a bigger role than genes
MZ=DZ

Concordance vastly different between MZ and DZ shows **genes** play a bigger role than environment
MZ >>DZ



From Medical Epigenetics 2021

The greater the difference in concordance between MZ and DZ twins = **greater genetic input**
MZ >>DZ

- MZ = Identical twins, same genome
- DZ = Fraternal twins, genome similar like siblings (50%)

Measuring Genetic Contribution in Multifactorial Disorders

Limitations to twin studies

- Underestimate heritability (compares differences between 100% and 50% identical genomes)
- MZ twins may have some different genes (e.g. mitochondrial genes) and epigenetic differences, and somatic mutations due to mitotic errors
- Different environmental exposures (*in utero* and after birth)
- Different genes in different twin pairs (studies between different sets of twins may point to different contributors for same phenotype)
- Ascertainment bias (when collecting data)
- Most studies do NOT specify the loci and alleles but how genotype and environment interact

Measuring Genetic Contribution in Multifactorial Disorders

Other methods

Family studies

- Similar to twin studies but not as precise
- More genetic variability

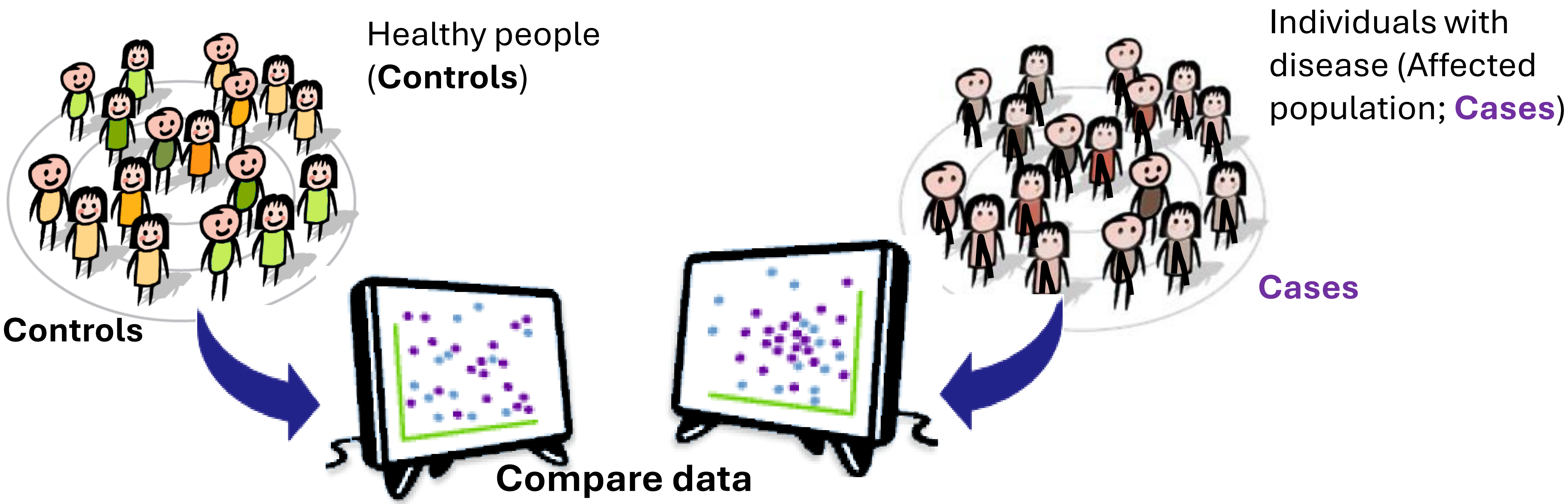
Adoption studies

- Separation of identical twins = same genes but different environment
- Can yield good data, but logistically very difficult

Identifying Gene Susceptibility Loci in Multifactorial Disorders

Genome-Wide Association Study (GWAS)

Testing co-occurrence of a specific allele at a marker locus and a trait in a population by comparing frequency of an allele in patients vs. controls

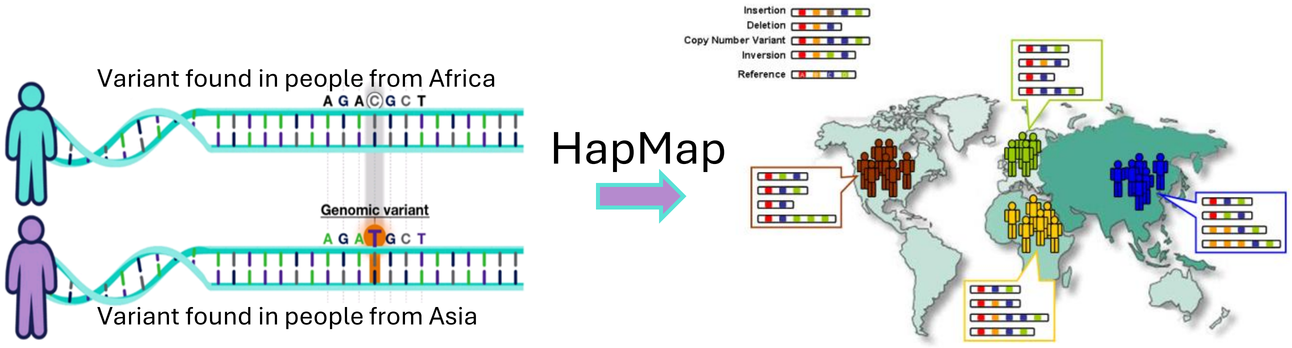


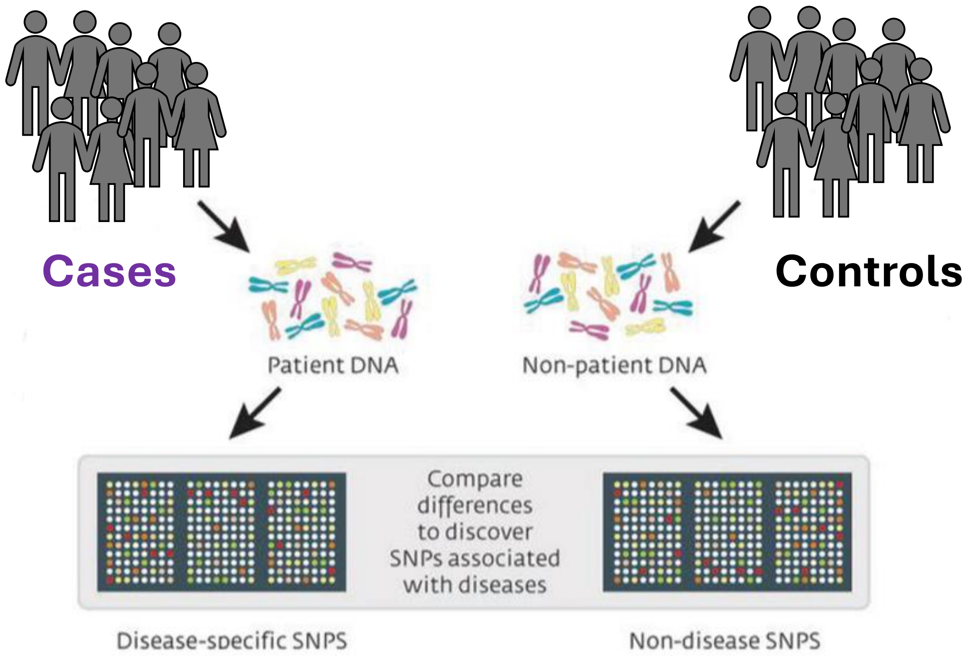
- **Genetic markers** are studied that are more commonly found in Affected vs. Unaffected
- Known as "**case-control studies**"
- These disorders do not follow familial inheritance patterns

Identifying Gene Susceptibility Loci in Multifactorial Disorders

Genome-Wide Association Study (GWAS)

- Association studies meet microarray
- HapMap project identified ~500,000 common SNPs and made a microarray
- GWAS compares genomes of affected and unaffected (controls) individuals
- Associations identified between SNPs and phenotype





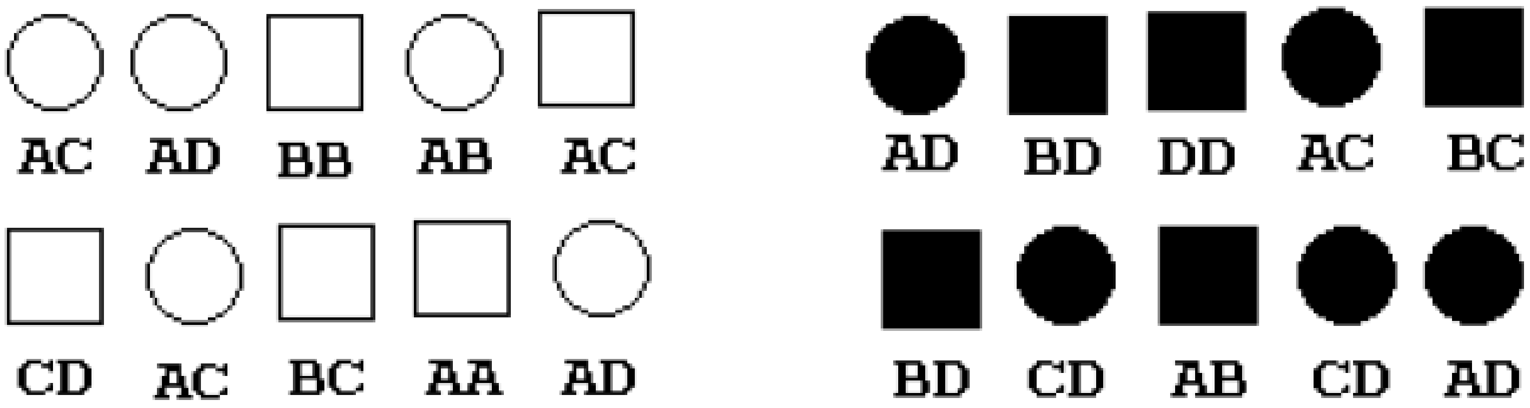
Identifying Gene Susceptibility Loci in Multifactorial Disorders

Genome-Wide Association Study (GWAS)

Frequency of alleles at a particular locus:

Controls — **Cases (Affected Disorder)**

‘D’ found 3 times — **‘D’ found 8 times**

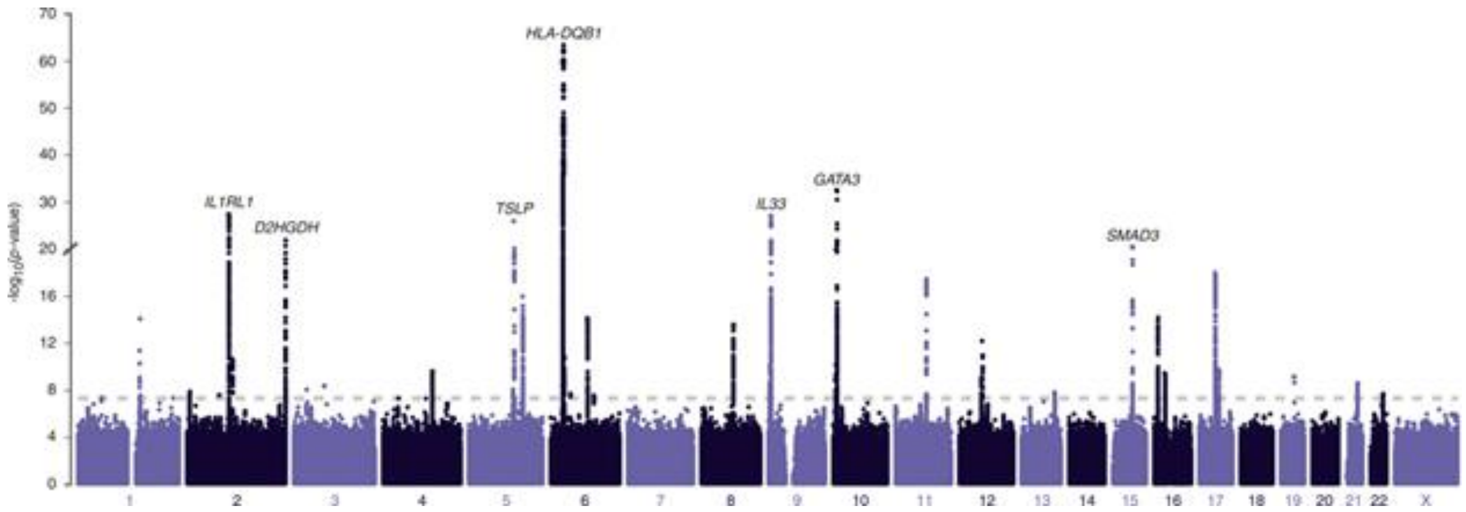


- Allele ‘**D**’ found more often in affected individuals than in healthy population (controls)
- Therefore, **D** allele is possibly a disease-causing mutation (or a predisposing risk factor),
- Or linked to disease-causing locus
- Caveat: Association studies give only a correlation between presence of a marker and the occurrence of disease

Identifying Gene Susceptibility Loci in Multifactorial Disorders

Known example: GWAS for Susceptibility for Asthma

- Below is an actual data for GWAS of asthma
- Compares genotypes at millions of genetic variants between asthma cases and controls
- Genotypes associated with asthma are shown above threshold
- The more associated signals a patient has above the threshold, greater risk for asthma



Threshold

https://www.ebi.ac.uk/gwas/downloads/summary-statistics

Many more examples

Role of Environmental factors in Multifactorial Disorders

- Population/migration studies
- Adoption studies (not discussed)

Evidence for Environment in Multifactorial Disorders

Population/Migration studies

Australia is very sunny, and population is fair skinned; increased risk for skin cancer

People in Japan eat lot of fish, increases risk of stomach cancer (involves chemistry of nitrogen compounds)

Countries showing highest and lowest incidence of specific types of cancer[a]			
Cancer site	Country of highest risk	Country of lowest risk	Relative risk H/L[b]
Skin (melanoma)	Australia (Queensland)	Japan	155
Lip	Canada (Newfoundland)	Japan	151
Nasopharynx	Hong Kong	United Kingdom	100
Prostate	U.S. (African American)	China	70
Liver	China (Shanghai)	Canada (Nova Scotia)	49
Penis	Brazil	Israel (Ashkenazic)	42
Cervix (uterus)	Brazil	Israel (non-Jews)	28
Stomach	Japan	Kuwait	22
Lung	U.S. (Louisiana, African American)	India (Madras)	19
Pancreas	U.S. (Los Angeles, Korean American)	India	11
Ovary	New Zealand (Polynesian)	Kuwait	8
Lung, male	Eastern Europe	West Africa	33
Esophagus	Southern Africa	West Africa	16
Colon, male	Australia, New Zealand	Middle Africa	15
Breast, female	Northern Europe	China	6

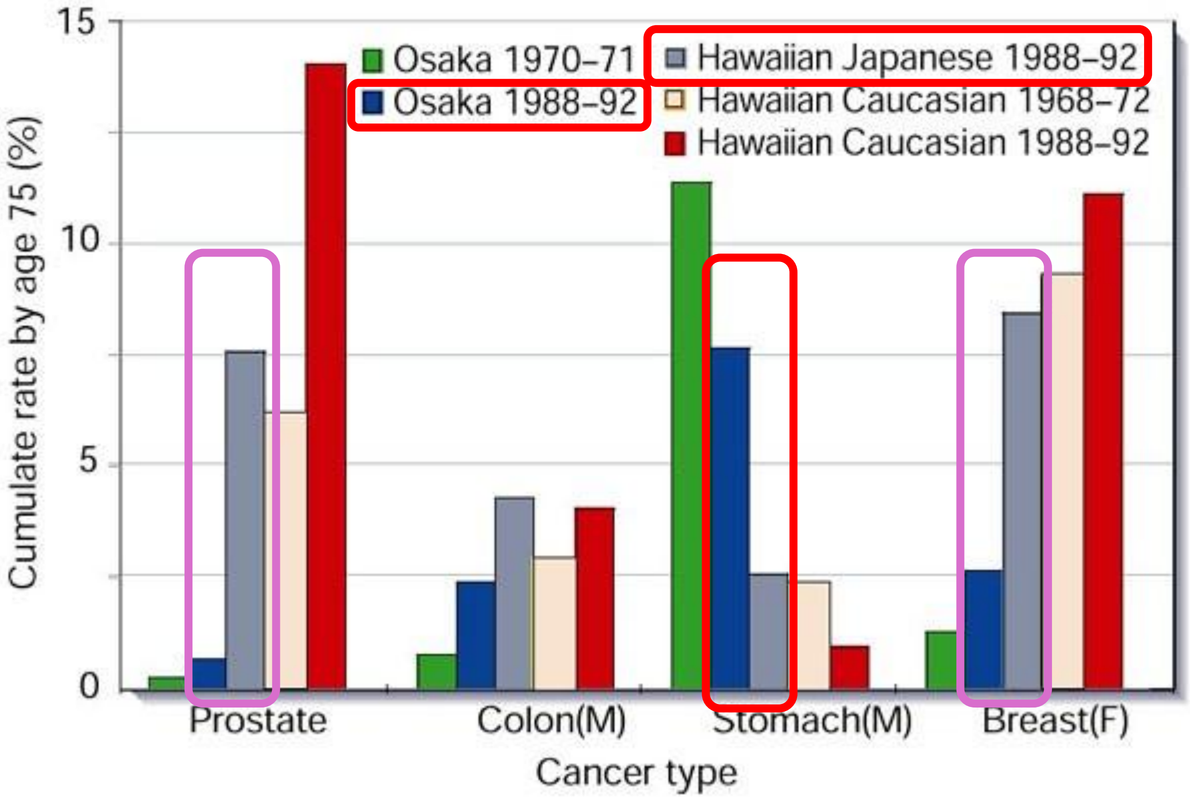
Table 2.5 part 1 of 2 *The Biology of Cancer* (© Garland Science 2007)

Evidence for Environment in Multifactorial Disorders

Example: Population/Migration studies

Japanese people studied

- Japanese living in Hawaii have higher incidence of **prostate and breast cancer** vs. Japanese living in Japan
  - **Role of Environmental/ cultural factors** as genetics are similar
- Japanese living in Japan have higher incidence of **stomach cancer** than Japanese living in Hawaii
  - **Role of environmental/ cultural factors** as genetics are similar



J. Peto, 2001

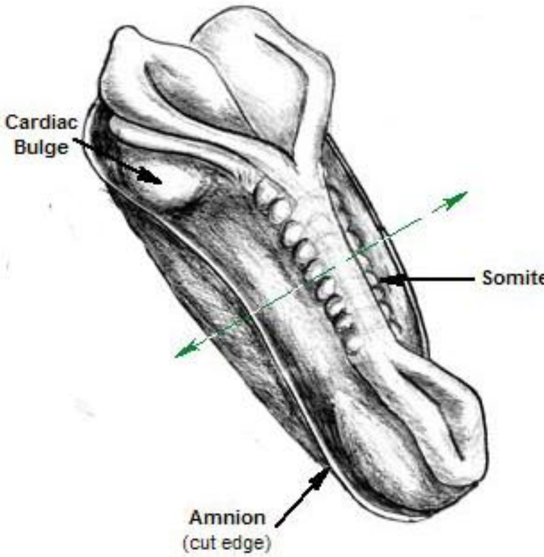
Environment & Genetic Contributors in Multifactorial Disorders

Known Example: Neural tube defects

Anencephaly

- Most severe of the neural tube defects
- Rare and fatal
- Underdeveloped brains and incomplete skulls
- May have brain stem function
- Most do not survive more than a few hours after birth

Anencephaly



Spina bifida (1/1000 live births)

- Incomplete closure of the spine
- Can be quite variable in severity
- Repair may sometimes be done in utero or postnatally, but successful outcomes are variable

Spina bifida

Both show multifactorial inheritance

Environment & Genetic Contributors in Multifactorial Disorders

Known Example: Neural tube defects

Evidence for genetics

- Varying prevalence in different human populations
- A woman who has had one child with neural tube defect such as spina bifida has ~3% risk of having another (*much higher than population risk*)
- Mutations in genes coding enzymes for folate metabolism (**MTFHR**)

Evidence for environment

- Maternal **dietary folate supplementation** reduces incidence of neural tube defects by about 70%
- So, the 3% risk can be reduced to 1% with high doses of folic acid

Environment & Genetic Contributors in Multifactorial Disorders
Known Example: Alzheimer's Disease, progressive dementia

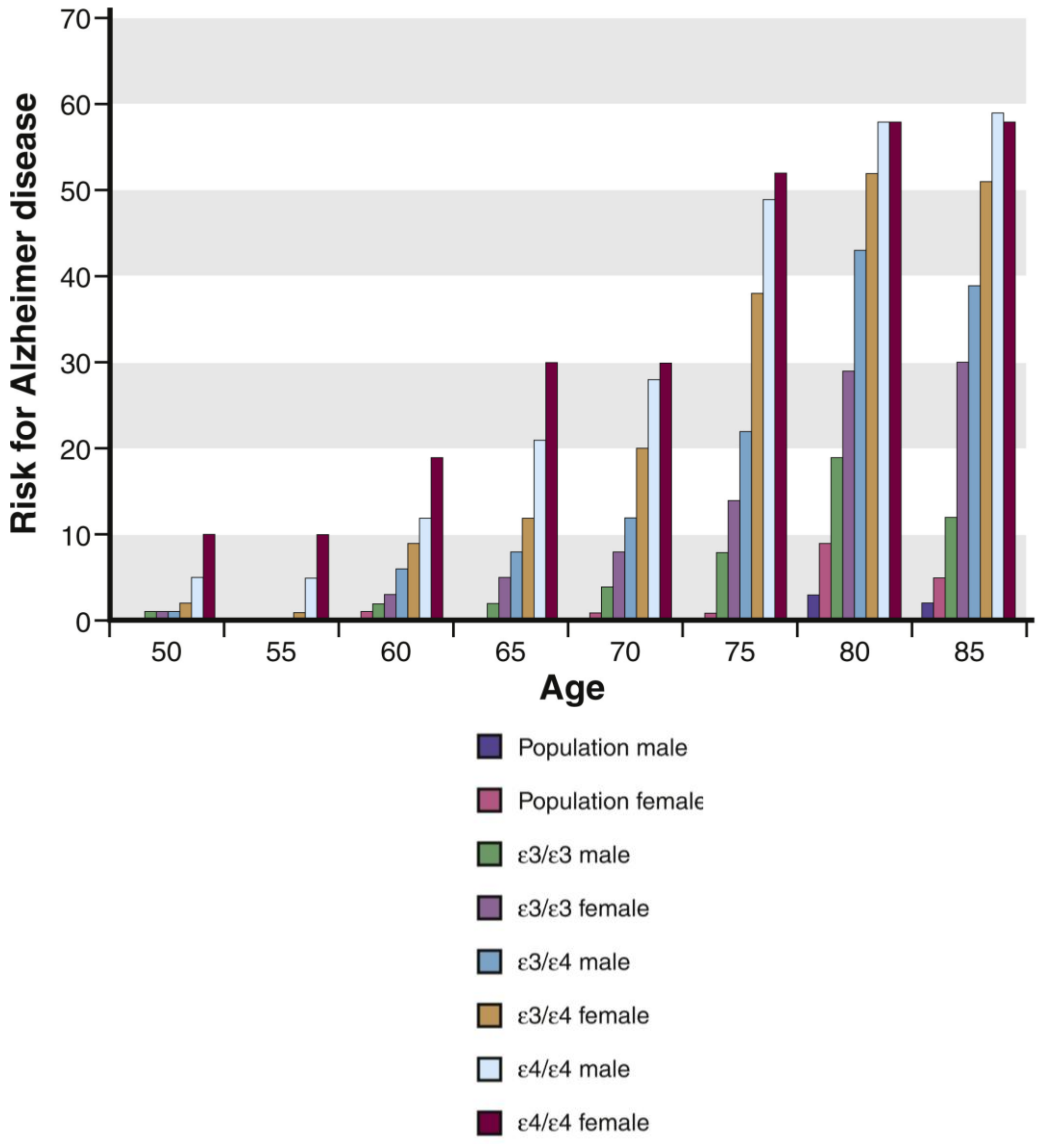
Diagnosis is established clinically

Evidence for genetics

- **Late onset Alzheimer's, with three common alleles**
  - Apolipoprotein E (ApoE) alleles ε2, ε3, & **ε4**
    - **ApoEε4 allele** in affected 3X than in controls (**Susceptibility** allele)

Evidence for environment

- Advanced age is a risk
- Environmental exposure
  - Oxidative damage to cells and mitochondria → Neuroinflammation and apoptosis
  - **Pesticide** exposure
  - Industrial **chemical** exposure such as flame retardants
  - Exposure to **toxic metals** (copper, lead and manganese)



Study guide

- Compare incidence of multifactorial diseases to single gene disorders
  - Recognize the clinical importance of multifactorial disorders
- Distinguish characteristics of multifactorial inheritance from Mendelian inheritance
  - Differentiate between quantitative traits and threshold traits
  - Outline polygenic theory of quantitative traits
- Explain how both genetic and environmental factors contribute to quantitative traits
- Outline the liability/threshold model of multifactorial inheritance
  - Compare and contrast quantitative traits to threshold traits
  - How bias may shift the threshold
- Describe approaches on how genetic and environmental factors involved in multifactorial disorders are separated (use of twin studies, population migration studies)
- Interpret heritability of quantitative traits (concordance and discordance)
- Outline how twin studies have help to understand heritability
- Discuss the effect of differences in incidence based on the sex of the individual for disorders that show differing liability for males and females

Thank you!

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